10. MCP

MCP

Model Context Protocol

- open protocal that standardizes how AI applications connect to tools, data, resources, and services.

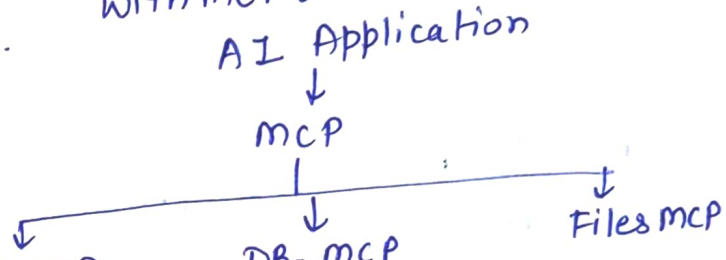
MCP = standard interface for AI applications to discover & use external capabilities.

Why MCP?

Without MCP, every AI application may need seperate integrations:

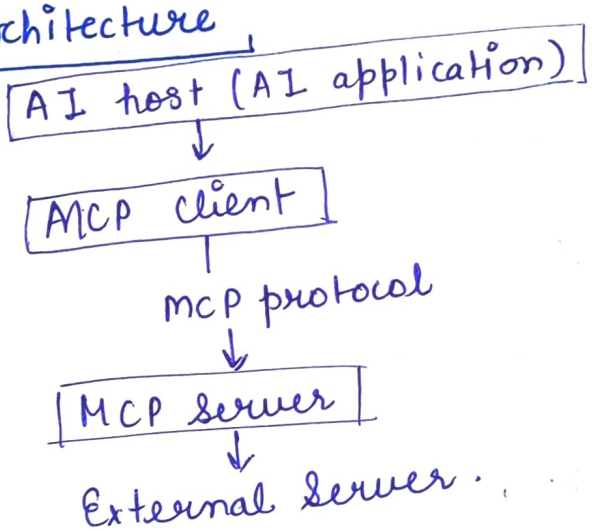
AI app
- custom Github integration.
- " DB "
- " Slack "
- " Drive "
- " Weather "

With MCP:



Main benefit: Standardized + reusable integration.

MCP Architecture



Remeber:
Host → Client → → Server → → External Server.

MCP host

Host = AI application that wants to use external capabilities.

User → AI host.

MCP Client

client = maintains connection b/w AI host & MCP server.

AI host → MCP client → MCP server.

client connects to the server.

MCP Server

server = program that implements MCP & exposes capabilities.

MCP server
- └ search files
- └ Real files
- └ Query Database
- └ create records.

It connects exposed capabilities to the underlying system.

MCP Resources *

Resource = Information | data made available to the AI application.

Examples : Documents, Files, Database records, API data, Application Information.

MCP server → Company Documentation → AI application.

MCP Tools

Tool = action the AI application can request.

Examples → search_files(), query_database(), create_ticket(), send_message(), get_weather()

Resource = data | Tool = action.

MCP Prompts

Prompt = reusable prompt template exposed through MCP.

MCP server → Prompt : "Review this code for security issue".

Three major capabilities :
- ⊙ Resources → Access information
- ⊙ Tools → Perform actions
- ⊙ Prompts → Reusable prompt templates.

Resources v/s Tools v/s Prompts

Component	Purpose	example
⊙ Resource	— Provides data/info	— Repository info
⊙ Tool	— Perform an action	— Create Github issue
⊙ Prompt	— Reusable promps / template.	— Review this repo.

MCP Communication Flow

User → AI app → MCP Client → MCP server → DB →

→ MCP server → MCP client → AI application → User.

Tool Directory, Discovery

→ MCP allows the Client to discover capabilities exposed by a server.

MCP server → Available tools : search_documents
get_customer
create_ticket

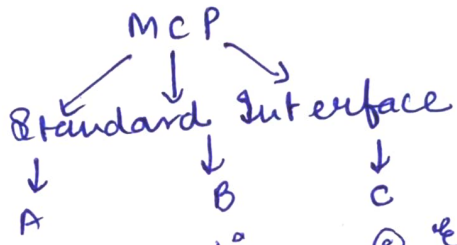
The application can then select the relevant capability.

Tool discovery = find available capabilities before using them.

Why Standardization

without MCP : AI ⟷ custom integration A
AI ⟷ " B
" " C

with MCP :



Behefits : ⊙ Consistent integration
⊙ Reusable MCP servers
⊙ Easier expansion.
⊙ Easier tool discovery
⊙ less custom integration work

MCP + AI agents

An agent can use MCP when it ~~to~~ needs external information or an action.

User request → Agent → need external information/action →
→ MCP tool → Result → Agent Continues #

Eg : Agent → MCP Github server → Repo.Tool → Issue Infomath → Agent → As. (Answer)

MCP v/s API

API : Application → Github API → Github

MCP : AI Application → MCP → MCP server → Github API → Github.

MCP doesn't replace APIs. An MCP server can use APIs internally.

MCP v/s Tool calling

Tool calling : Agent → Tool

MCP : Agent → MCP client → MCP server → Tool/Resource

⊙ Tool calling ⇒ capability
⊙ MCP ⇒ standardized Protocol for exposing & using capabilities.

MCP V/s A2A

A2A : Agent A ⟷ Agent B # used for agent Collaboration

MCP : AI agent → MCP → Tool | Data | service.

used for connecting AI applications to external capabilities & information.

A2A # Agent ⟷ Agent MCP # AI Application ⟷ Tools | Data | services

MCP in Agent Workflow

User
↓
Agent
↓
Orchestrator
↓
Agent A MCP Client
↓
MCP server
↓
Extraction Tool
↓
Result
↓
Agent.

Orchestration → controls workflow
A2A → agents communicate
MCP → connects AI to external tools | data.

Security in MCP

Important controls : Authentication • Authorization • Permissions • Input validation • Access Control • Data Privacy • Secure Communication.

Eg : Agent → MCP server → "Delete customer Database".

Powerful actions should require appropriate permission/control.

Least Privilege

Give tools/servers only the permissions they need.

✓ Read files × Delete, Modify. Access unrelated systems.

Least Privilege = min. permissions required to perform the task.

MCP Example — Coding Assitant

AI coding Assistant
↓
MCP client
↓
MCP server
↙ ↓ ↘
Files Github Database

Possible capabilities :

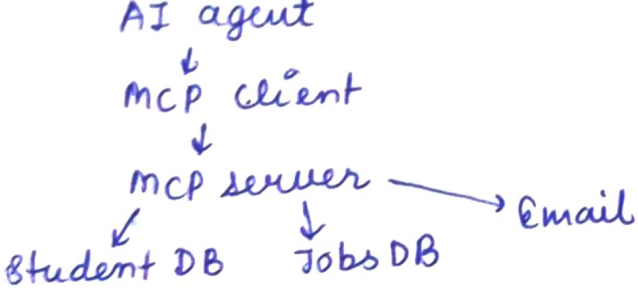
Resource ⇒ Project Files

Tool ⇒ Create Github issue

Tool ⇒ Run tests

Resource ⇒ Repo. Info

MCP Example - Student Platform



Possible capabilities :

Resource - Student Profile

Resource - Internship listings

Tool - Search Internships

Tool - Send Notification.

MCP Workflow

AI application connects to MCP server → Server enposes available capabilities.

→ App. discovers tools/resources → Agent decides what is needs →

→ MCP req. is sent → Server perfoms / retrieves operation →

→ Result is returned → Agent Continues the task.

Must Remember

- MCP = Model Context Protocol
- Standardized way for AI apps. to connect to tools / data / services.
- Architecture : Host → Client → Server → External System.
- Resource : data , Tool : action.
- Prompt : reusable prompt / template.
- Tool discovery allows available capabilities to be discovered.
- MCP doesn't replace APIs
- Security : Auth + Authorization + Least Privilege + Validation.
- MCP is especially useful in AI agents & orchestrated workflow

MCP = Standardized protocol that lets AI application disover & interact with external ~~tools~~ tools, data, and services.

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